

Linear Programming

1 Day 1: Foundations of Linear Programming

What is linear programming (LP)? It is a **mathematical optimization framework** for making decisions under constraints.

1.1 Formulating Linear Programming Problems

Kaiser and Messer emphasize that formulation is the central task in LP: the analyst translates a verbal decision problem into a mathematical model. Begin by reading the problem carefully to understand the decision setting before trying to write equations. Then read it again to sort the information needed for the model.

1.1.1 A Practical Formulation Sequence

1. **Identify the objective.** What is the decision maker trying to maximize or minimize—for example, profit, cost, sales, or production?
2. **Identify the activities (decision variables).** What choices can the decision maker make to pursue the objective? Define the unit of measurement for each activity.
3. **Identify objective coefficients.** For each activity, determine its per-unit contribution to the objective and record its units.
4. **Identify resources or requirements.** List the resources controlled by the decision maker, their available amounts, and their units. In problems without fixed resources, identify the relevant right-hand-side requirements instead.
5. **Identify technical coefficients.** For every activity-resource pair, determine how much of the resource is used or supplied by one unit of the activity. Check the units carefully.
6. **Write the structural constraints.** Translate resource limits or minimum requirements into $\leq$, $\geq$, or $=$ constraints.
7. **Add variable restrictions and assemble the model.** Include nonnegativity (or other sign restrictions) and combine the objective and constraints into a complete scalar model. Descriptive constraint labels make the model easier to audit and interpret.

Throughout the process, check that every term within an equation has compatible units. Formulation improves through practice: solve many verbal problems, compare alternative specifications, and ask whether each equation faithfully represents the decision setting.

1.2 Farm LP Example

A farmer has 500 acres of land available and is deciding how to allocate it between wheat and corn. Each acre of wheat yields a profit of \$200, requires 3 hours of labor, and 4 units of fertilizer. Each acre of corn yields a profit of \$300, requires 4 hours of labor, and 3 units of fertilizer. The farm has at most 1,800 hours of labor available and 2,000 units of fertilizer.

Formulate this situation as a linear programming problem. Clearly define the decision variables, write down the objective function representing total profit, and specify the constraints that capture the land, labor, fertilizer, and nonnegativity restrictions.

Decision variables.

Let W = acres of wheat, C = acres of corn.

1.2.1 Scalar form

$$\begin{aligned} \max_{W,C} \quad & 200W + 300C \\ \text{s.t.} \quad & W + C \leq 500 \quad (\text{land}) \\ & 3W + 4C \leq 1800 \quad (\text{labor}) \\ & 4W + 3C \leq 2000 \quad (\text{fertilizer}) \\ & W, C \geq 0. \end{aligned}$$

1.2.2 Matrix (compact) form

$$\begin{aligned} \max_{x \in \mathbb{R}_{\geq 0}^2} \quad & c^\top x \\ \text{s.t.} \quad & Ax \leq b, \end{aligned} \quad \text{with} \quad x = \begin{bmatrix} W \\ C \end{bmatrix}, c = \begin{bmatrix} 200 \\ 300 \end{bmatrix}, A = \begin{bmatrix} 1 & 1 \\ 3 & 4 \\ 4 & 3 \end{bmatrix}, b = \begin{bmatrix} 500 \\ 1800 \\ 2000 \end{bmatrix}.$$

1.3 Properties of LP

- **Objective** is to either maximize or minimize
- **Constraints:** linear inequalities or equalities that limit the feasible choices (e.g., land, labor, fertilizer).
- All equations are **linear**
- **Decision variables:** the choices to be made (e.g., how many acres of wheat and corn to plant) are non-negative.

1.4 Assumptions of Linear Programming

The first three assumptions concern whether the model represents the decision problem appropriately; the remaining four state its mathematical and information requirements.

- **Objective-function appropriateness:** one stated objective is sufficient to rank feasible choices.
- **Decision-variable appropriateness:** all relevant choices are included and controlled by the decision maker.
- **Constraint appropriateness:** the constraints accurately represent the relevant limits, requirements, and units.
- **Proportionality:** each activity's per-unit contribution and resource use do not change with its level.
- **Additivity:** the contribution of one activity does not depend on the level of another activity.
- **Divisibility:** decision variables may take fractional values.
- **Certainty:** objective, technical, and right-hand-side coefficients are known fixed values.

When an assumption is unsuitable, the model may need an extension: multi-objective programming, integer programming, nonlinear programming, or stochastic programming.

1.5 Graphical Solution Method (Two Variables)

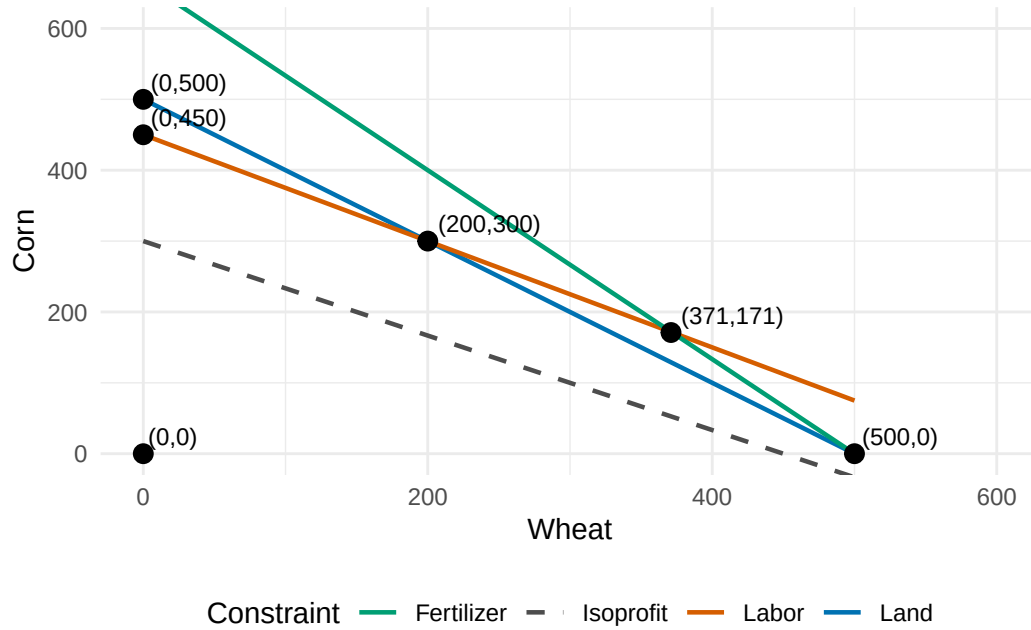
1.5.1 Key Geometry Terms

- A **feasible solution** satisfies every constraint; an **infeasible solution** violates at least one.
- The **feasible region** is the set of all feasible solutions.
- The feasible region of an LP is **convex**: the line segment joining any two feasible solutions is also feasible.
- An **extreme point** (or vertex) is a corner of the feasible region.
- A constraint is **binding** when it holds with equality at a solution; otherwise, it has **slack**.

Step 1. Draw the constraints.

- Land: $W + C \leq 500$
- Labor: $3W + 4C \leq 1800$
- Fertilizer: $4W + 3C \leq 2000$
- Nonnegativity: $W, C \geq 0$
- Plot based on endpoints. Set one variable to zero and devote all resources to the other variable.

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Step 2. Identify the feasible region.

- Intersection of all constraints in the (W, C) plane.
- Polygon bounded by lines.

Step 3. Plot the objective function.

- Profit = $200W + 300C$.
- Show isoprofit lines:
 - Suppose we plot \$90,000 profit. All wheat no corn, then all corn no wheat.
 - Lines of constant profit have slope $-\frac{200}{300} = -\frac{2}{3}$.

Step 4. Locate the optimum.

- Slide the isoprofit line outward until the last point of contact with the feasible region.
- Optimum is always at a **corner point** (fundamental theorem of LP).

Step 5. Verify algebraically.

Use the binding constraints to recover the exact solution, then check every constraint and calculate the objective value. In the farm example, the graphical optimum has $W = 0$ and the labor constraint binds:

$$4C = 1800 \Rightarrow C = 450.$$

The solution $(W, C) = (0, 450)$ satisfies land use ($450 \leq 500$) and fertilizer use ($1350 \leq 2000$), and it yields profit $200(0) + 300(450) = 135,000$.

1.6 Day 1 Summary

- LP is a framework for making decisions subject to constraints.
- Formulate an LP by identifying the objective, activities and their units, coefficients, resources or requirements, and variable restrictions.
- A standard LP combines a linear objective, linear structural constraints, and sign restrictions; its coefficients are summarized in c , A , and b .
- LP relies on assumptions about appropriate objectives, variables, and constraints, along with proportionality, additivity, divisibility, and certainty.
- In two-variable problems, graph the constraints to find the feasible region, use an isoprofit line to locate an extreme-point optimum, and verify the result algebraically.

2 Day 2: Solving Linear Programs

2.1 Graphical solution method (2 variables)

Vertex	Profit
$(W, C) = (0, 0)$	\$0
$(0, 450)$	\$135,000
$(200, 300)$	\$130,000
$(500, 0)$	\$100,000

- Why don't we need to check (371,171)? Because it is not a vertex of the feasible region.

Fundamental theorem of linear programming: If an LP has an optimal solution, at least one of them occurs at a vertex of the feasible region.

- Which constraints bind at the optimum?
- What parameters of the problem would need to change for wheat to enter the solution?

2.1.1 Carbon abatement problem: algebraic solution

Write the carbon-abatement problem in matrix form. Identify the decision variables, objective coefficients, technical coefficients, and right-hand-side requirements.

- The carbon-abatement model has three decision variables.

Show 3d plot of carbon-abatement feasible region.

- We can find a vertex by setting **three independent constraints** to equality.
- The candidate binding constraints can be emissions requirements or nonnegativity restrictions such as $B = 0$.

For example, set the nitrous-oxide requirement and $M = F = 0$ to equality:

$$5B = 75 \Rightarrow (B, M, F) = (15, 0, 0).$$

This candidate is feasible because it reduces carbon dioxide by $150 \geq 100$, methane by $30 \geq 25$, and nitrous oxide by 75. Its cost is $0.5(15) = \$7.5$ million.

To find another candidate, set the carbon-dioxide and nitrous-oxide requirements to equality and set $F = 0$:

$$\begin{aligned} 10B + 15M &= 100, \\ 5B + 12M &= 75. \end{aligned}$$

- Using Gaussian elimination, we can solve this system of equations. Doubling the second equation and subtracting the first gives $9M = 50$, so $M = 50/9$ and $B = 5/3$. This point is feasible because its methane reduction is $280/9 \approx 31.1 \geq 25$.

Repeating this process for all combinations of three binding constraints, and discarding any intersection that violates a remaining constraint, gives the feasible vertices below.

Binding constraints	(B, M, F)	Cost (million \$)
$\text{CO}_2, \text{N}_2\text{O}, B = 0$	$(0, 115/21, 5/7)$	8.93
$\text{CO}_2, \text{N}_2\text{O}, F = 0$	$(5/3, 50/9, 0)$	7.50

Binding constraints	(B, M, F)	Cost (million \$)
$\text{CO}_2, B = 0, F = 0$	$(0, 20/3, 0)$	8.00
$\text{N}_2\text{O}, B = 0, M = 0$	$(0, 0, 75/13)$	19.04
$\text{N}_2\text{O}, M = 0, F = 0$	$(15, 0, 0)$	7.50

The smallest cost is \$7.5 million. Because two vertices tie, every feasible point on the line segment between $(5/3, 50/9, 0)$ and $(15, 0, 0)$ is also optimal.

This is an example of **multiple optimal solutions**.

2.2 Simplex Method

2.2.1 Why We Need It

- The graphical method only works for **two or three variables**. Algebraic methods are tedious for large problems.
- Real problems may involve **hundreds or thousands** of variables.
- The simplex method provides a **systematic way** to move from vertex to vertex until the best one is found.

2.2.2 Core Idea

- Start at a **basic feasible solution (BFS)** — a corner point of the feasible region.
- At each step:
 1. Identify an **entering variable**: an activity whose introduction can improve the objective.
 2. Determine which current activity must decrease to keep the solution feasible; this is the **leaving variable**.
 3. **Pivot** to a neighboring BFS.
- Stop when no variable can improve the objective — this is optimal.

2.2.3 One pivot

Return to the wheat–corn model from Day 1. We can use it to see what the simplex terms mean without carrying out a long sequence of row operations:

$$\begin{aligned}
 \max \quad & 200W + 300C \\
 \text{s.t.} \quad & W + C \leq 500 \quad (\text{land}), \\
 & 3W + 4C \leq 1800 \quad (\text{labor}), \\
 & 4W + 3C \leq 2000 \quad (\text{fertilizer}), \\
 & W, C \geq 0.
 \end{aligned}$$

Add slack variables to write the resource constraints as equalities:

$$\begin{aligned}
 W + C + s_L &= 500, \\
 3W + 4C + s_K &= 1800, \\
 4W + 3C + s_F &= 2000, \\
 s_L, s_K, s_F &\geq 0.
 \end{aligned}$$

At the origin, $W = C = 0$, so $(s_L, s_K, s_F) = (500, 1800, 2000)$. This is a BFS: the slack variables form the initial **basis** and record the unused resources.

- At this point, a unit of corn raises profit by \$300, compared with \$200 for a unit of wheat. Choose C as the **entering variable**.

- Let $C = t$ while $W = 0$. The slack equations become $s_L = 500 - t$, $s_K = 1800 - 4t$, and $s_F = 2000 - 3t$. Feasibility requires

$$t \leq 500, \quad t \leq \frac{1800}{4} = 450, \quad t \leq \frac{2000}{3} \approx 666.7.$$

- Labor is reached first, so s_K is the **leaving variable**. This is the **ratio test**: the smallest nonnegative limit determines how far the entering activity can increase.
- The pivot moves us to the neighboring vertex $(W, C) = (0, 450)$, where $(s_L, s_K, s_F) = (500, 0, 650)$ and $z = \$135,000$. The basis changes from $\{s_L, s_K, s_F\}$ to $\{s_L, C, s_F\}$.

Can we improve profit by adding wheat?

- To remain on the binding labor constraint, one additional acre of wheat requires reducing corn by 3/4 acre (ratio of technical coefficients). The resulting change in profit is

$$200 - 300 \left(\frac{3}{4} \right) = -25.$$

Thus wheat has no profitable direction in which to enter, and the current BFS is optimal. The simplex algorithm performs this same check-and-move process in higher dimensions, using matrix operations rather than a graph.

2.2.4 Intuition

- Simplex is like **walking along the edges** of the feasible polygon.
- At each corner, ask: *“If I move along this edge, does profit go up?”*
- Continue until no edge yields improvement.
- The same logic applies in higher dimensions, even though we cannot draw the polytope.

2.2.5 Optional Reference for Simplex Mechanics

For a matrix-algebra treatment of the simplex algorithm, see McCarl and Spreen, Chapter III, especially Sections 3.2 and 3.4. The graphical method above supplies the intuition needed to interpret what the algorithm is doing.

2.2.6 Excel Solver

Excel implements the simplex method in the solver add-on. See [LP_land_alloc.xlsx](#)

2.3 Day 2 Summary

- If a bounded LP has an optimum, at least one optimal solution occurs at a **vertex** of its feasible region.
- Simplex** generalizes the graphical corner-point search: it moves from one **basic feasible solution (BFS)** to a neighboring BFS until no feasible move improves the objective.
- Slack variables convert “ $\leq$ ” constraints to equalities, measure unused resources, and often provide a convenient initial basis.
- In each iteration, choose an **entering variable** with potential to improve the objective; the **ratio test** identifies the **leaving variable** and keeps the solution feasible; a **pivot** updates the basis.
- A different feasible starting vertex may change the path simplex takes, but not the optimal objective value.

3 Day 3: Sensitivity Analysis

3.1 Review

- LPs make 7 assumptions. What are a few?
- How do LPs algorithms solve? What is the fundamental theorem of LP?

3.2 Motivation

- LPs give an **optimal solution** and an **objective value**.
 - But: parameters (profits, resources, input requirements) are rarely known with certainty.
 - We need to know how **robust** the solution is.
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3.3 What Is Sensitivity Analysis?

- Also called **post-optimality analysis**.
- Asks how changes in model parameters affect the optimal activities and objective value.
- It supports both **what-if analysis** and, when a price varies systematically, a **price-quantity schedule** such as an individual supply curve.
- Focus on three categories:
 1. Objective coefficients (c_j : unit profits, costs, or prices),
 2. RHS values (b_i : resource endowments or requirements), and
 3. Technical coefficients (a_{ij} : input-output relationships).

The objective-coefficient walkthrough below follows the sequence in Kaiser and Messer, Chapter 1, pp. 34–37, but applies every step to our wheat–corn farm model.

3.4 Walk-Through: Objective-Coefficient Sensitivity

Returning to the wheat–corn farm example, the LP is:

$$\begin{aligned} \max_{W,C} \quad & Z = 200W + 300C \\ \text{s.t.} \quad & W + C \leq 500 && (\text{land}) \\ & 3W + 4C \leq 1800 && (\text{labor}) \\ & 4W + 3C \leq 2000 && (\text{fertilizer}) \\ & W, C \geq 0. \end{aligned}$$

3.4.1 Step 1: Start at the baseline optimum

The baseline solution is $W^* = 0$, $C^* = 450$, and $Z^* = \$135,000$. Labor is binding; land and fertilizer have slack. The point also lies on the nonnegativity boundary $W = 0$.

Changing an objective coefficient changes the direction of an iso-profit line, but it does not move the feasible region. The question is therefore: how far can that line rotate before another corner becomes better than the corn-only corner?

3.4.2 Step 2: Write the relevant slopes

Use general unit-profit coefficients c_W and c_C :

$$\max Z = c_W W + c_C C.$$

Solving an iso-profit line for corn gives

$$C = \frac{Z}{c_C} - \frac{c_W}{c_C}W,$$

so its slope is $-c_W/c_C$. The labor constraint, which is the feasible edge leaving the corn-only corner, is

$$C = 450 - \frac{3}{4}W,$$

with slope $-3/4$.

For the corn-only corner to remain optimal, moving rightward along the labor boundary must not increase profit. Thus, the iso-profit line must be no steeper than that edge:

$$-\frac{c_W}{c_C} \geq -\frac{3}{4} \quad \Longleftrightarrow \quad \frac{c_W}{c_C} \leq \frac{3}{4}.$$

Unlike the reading's feed-dealer example, this corner has only one adjacent structural constraint edge; the other binding condition is $W = 0$. Therefore the coefficient ranges below have one finite endpoint rather than two.

3.4.3 Step 3: Range for wheat's unit profit

Hold corn's unit profit at $c_C = 300$. Then

$$\frac{c_W}{300} \leq \frac{3}{4} \quad \Longrightarrow \quad c_W \leq 225.$$

So wheat's unit profit can rise from \$200 to \$225 per acre without changing the unique corn-only optimum. At $c_W = 225$, the labor edge has the same slope as the iso-profit line: every feasible mix on that edge from $(W, C) = (0, 450)$ to $(200, 300)$ is optimal. Above \$225, wheat enters the optimal plan.

There is no finite allowable decrease for wheat's objective coefficient in this example: making an activity that is already unused even less attractive does not displace corn.

3.4.4 Step 4: Range for corn's unit profit

Hold wheat's unit profit at $c_W = 200$. Then

$$\frac{200}{c_C} \leq \frac{3}{4} \quad \Longrightarrow \quad c_C \geq 266.67.$$

Corn's unit profit can fall from \$300 to \$266.67 per acre without changing the unique corn-only optimum. At $c_C = 266.67$, the same alternative-optima edge appears; below that level, wheat enters the plan. There is no finite allowable increase for corn's unit profit, because a higher corn return reinforces the corn-only solution.

3.4.5 Step 5: Separate activity levels from the objective value

Within the range, the activity levels stay at $W = 0$ and $C = 450$, but profit changes with the coefficient. For example, if corn's unit profit rises to \$320, the plan remains corn-only while profit becomes

$$Z^* = 320(450) = \$144,000.$$

This is the central distinction in objective-coefficient ranging: the **optimal basis and activity levels** can remain unchanged even though the **optimal objective value** changes.

Discussion prompts

1. If wheat profit rises to \$220, should we re-solve the LP? Why or why not?
2. Why is \$225 a threshold for alternative optima rather than a point at which a solver must choose a particular positive wheat acreage?
3. If both crop prices change, why must we check the ratio c_W/c_C rather than apply two one-at-a-time ranges independently?

3.5 Walk-Through: Resource Endowment (RHS) Sensitivity

This walkthrough follows the resource-endowment method in Kaiser and Messer, Chapter 1, pp. 38–40, but applies it to the wheat–corn farm. Hold the objective coefficients and all other model parameters constant while changing one RHS value at a time.

3.5.1 Step 1: Identify resource use at the baseline optimum

The three resource endowments are

$$b_{\text{land}} = 500, \quad b_{\text{labor}} = 1800, \quad b_{\text{fert}} = 2000.$$

At $(W^*, C^*) = (0, 450)$, the farm uses 450 acres of land, 1,800 labor hours, and 1,350 units of fertilizer. Labor is binding; land has 50 acres of slack and fertilizer has 650 units of slack.

Question. If the farm obtains one additional unit of a resource, how much does optimal profit change?

3.5.2 Step 2: Define the shadow price

Following Kaiser and Messer, change resource i by a small amount, re-solve the model, and compare the new and original optimal values:

$$SP_i = \frac{Z^*(b_i + \Delta b_i) - Z^*(b_i)}{\Delta b_i}.$$

For a one-unit increase, $\Delta b_i = 1$, so the shadow price is simply the change in optimal profit. Because an LP value function is piecewise linear, this marginal value is exact for any RHS change that remains within the allowable range.

3.5.3 Step 3: Increase the binding labor endowment by one hour

Raise labor availability from 1,800 to 1,801 hours. If the current basis remains optimal, wheat stays at zero and the new labor constraint binds:

$$3W + 4C = 1801, \quad W = 0.$$

Therefore,

$$C' = \frac{1801}{4} = 450.25.$$

Check the remaining constraints:

$$450.25 \leq 500 \quad (\text{land}), \quad 3(450.25) = 1350.75 \leq 2000 \quad (\text{fertilizer}).$$

The new solution is feasible and its profit is

$$Z' = 300(450.25) = \$135,075.$$

Thus,

$$SP_{\text{labor}} = \frac{135,075 - 135,000}{1801 - 1800} = \$75 \text{ per labor hour.}$$

One more labor hour permits one-quarter acre of additional corn, worth $300(1/4) = 75$ dollars. This is the operation's marginal value of labor capacity before paying to acquire it.

3.5.4 Step 4: Check the nonbinding resources

Increasing land from 500 to 501 acres leaves the plan at $(0, 450)$: land slack rises from 50 to 51 acres, while profit remains \$135,000. Therefore, $SP_{\text{land}} = 0$ at the current endowment.

Likewise, increasing fertilizer from 2,000 to 2,001 units only raises fertilizer slack. The activity levels and profit do not change, so $SP_{\text{fert}} = 0$ at the current endowment.

A zero shadow price does not mean that a resource can never have value. It means that one additional unit has no value **at the current endowment and basis**.

3.5.5 Step 5: Derive the allowable labor range

Let labor availability be the variable RHS b_{labor} . As long as the corn-only basis remains feasible,

$$C = \frac{b_{\text{labor}}}{4}, \quad s_{\text{land}} = 500 - \frac{b_{\text{labor}}}{4}, \quad s_{\text{fert}} = 2000 - \frac{3b_{\text{labor}}}{4}.$$

All basic variables must remain nonnegative. Ask what would b_{labor} have to be for each basic variable to become zero:

$$\begin{aligned} C \geq 0 &\implies b_{\text{labor}} \geq 0, \\ s_{\text{land}} \geq 0 &\implies b_{\text{labor}} \leq 2000, \\ s_{\text{fert}} \geq 0 &\implies b_{\text{labor}} \leq 2666.67. \end{aligned}$$

The tightest conditions give

$$0 \leq b_{\text{labor}} \leq 2000.$$

Starting from 1,800 hours, the allowable decrease is 1,800 hours and the allowable increase is 200 hours. Within this range,

$$C^* = \frac{b_{\text{labor}}}{4}, \quad Z^* = 75b_{\text{labor}},$$

so the shadow price remains \$75 per hour. At 2,000 hours, land also becomes binding. Beyond that breakpoint, the farm cannot expand corn acreage, so the marginal value of more labor falls to zero.

3.5.6 Step 6: Derive the ranges for land and fertilizer

At the baseline solution, corn uses 450 acres and 1,350 fertilizer units. Therefore:

- The current basis remains feasible for $b_{\text{land}} \geq 450$. The allowable decrease is 50 acres, the allowable increase is unbounded, and the current shadow price is zero.
- The current basis remains feasible for $b_{\text{fert}} \geq 1350$. The allowable decrease is 650 units, the allowable increase is unbounded, and the current shadow price is zero.

The one-at-a-time RHS sensitivity results are:

Resource	Current RHS	Use	Slack	Shadow price	Allowable decrease	Allowable increase
Land	500	450	50	\$0/acre	50	∞
Labor	1,800	1,800	0	\$75/hour	1,800	200
Fertilizer	2,000	1,350	650	\$0/unit	650	∞

3.5.7 Step 7: Use the ranges for what-if questions

- **Add 100 labor hours.** This is within the allowable increase. Corn rises to 475 acres and profit rises by $75(100) = 7,500$ dollars.
- **Add 250 labor hours.** This exceeds the allowable increase. Applying \$75 to all 250 hours would incorrectly predict an \$18,750 gain. Re-solving gives 500 corn acres and \$150,000 profit, a gain of only \$15,000, because land becomes binding after the first 200 additional labor hours.
- **Lose 75 acres of land.** The first 50 acres reduce slack but not profit. The remaining 25-acre loss forces corn down to 425 acres, reducing profit by $300(25) = 7,500$ dollars.

At an allowable-range endpoint, interpret the shadow price as one-sided: the marginal value on one side of the kink can differ from the marginal value on the other.

Discussion prompts

1. Why does labor have a positive shadow price while land and fertilizer initially have zero shadow prices?
2. Why can activity levels change even though the current basis and shadow price remain valid?
3. What resource becomes binding after labor rises above 2,000 hours, and how does that alter the value of additional labor?

3.6 Solver Connection: Reduced Costs

- The graphical condition above says that wheat can enter only when its unit profit exceeds \$225. In the Excel maximization convention, wheat's reduced cost is therefore $200 - 225 = -25$.
- A nonbasic activity has a reduced cost of zero at the entry threshold; at the threshold, alternative optima can occur. Sign conventions differ across solvers, so always interpret the reported value using the solver's documentation.
- Changing an objective coefficient never changes primal feasibility. For a basic activity, the solver checks whether the implied dual prices and reduced costs remain feasible.

3.7 Technical Coefficient Changes

- What if technology or input requirements change?
- A change in a_{ij} changes the feasible region itself, so it is not covered by the usual one-at-a-time objective-coefficient or RHS ranges in a Solver sensitivity report.

- For a small change that preserves the current basis, a shadow-price approximation is

$$\Delta Z^* \approx -y_i^* x_j^* \Delta a_{ij},$$

where y_i^* is the shadow price of resource i and x_j^* is the activity level.

- Example: if corn's labor requirement rises from 4 to 4.02 hours per acre, the wheat–corn model predicts $\Delta Z^* \approx -75(450)(0.02) = -\675 . Re-solve for a material change.
- If an activity is currently zero, changing one of its technical coefficients has no local effect on the objective value, although a sufficiently large change can make it enter.

3.8 Wheat–Corn Example (Excel Solver Report)

3.8.1 Excel Top Panel: Variable Cells

Interpretation of columns:

- **Final Value** – the optimal level of the decision variable (e.g., acres of wheat or corn).
- **Reduced Cost** – for nonbasic variables (value = 0), how much the objective coefficient must improve before the variable would enter the solution. Zero if the variable is positive in the solution.
- **Objective Coefficient** – the profit (or cost) per unit used in the objective function.
- **Allowable Increase / Decrease** – the range over which the objective coefficient can change without altering the current optimal basis (solution structure).

Interpretation of results

- Corn: optimal solution plants 450 acres of corn. Reduced cost = 0 because corn is already in the solution
- Wheat: optimal solution plants 0 acres of wheat. Reduced cost = -25 means a \$25/acre increase (from 200 to 225) reaches the alternative-optima threshold; an increase beyond \$25 makes wheat enter the solution.

Ranges:

- Wheat: profit can increase by 25 before the current corner ceases to be uniquely optimal.
- Corn: profit can fall as much as 33.3 (300 $\rightarrow$ 266.7) before solution changes.
- Interpretation: Corn dominates under current prices. Wheat only becomes attractive if its relative profit improves substantially.

3.8.2 Excel Bottom Panel: Constraints

Interpretation of columns:

- **Final Value** – the amount of the resource actually used at the solution.
- **Constraint R.H. Side** – the available amount of the resource (the right-hand side of the inequality).
- **Shadow Price** – the marginal value of relaxing the constraint (increase in objective if RHS increases by 1 unit), valid only within the allowable range.
- **Allowable Increase / Decrease** – the range over which the shadow price remains valid and the current basis stays optimal.

Interpretation of results:

- Land: only 450 acres used out of 500. Slack = 50 acres. Shadow price = 0 because land is not binding. You can increase land indefinitely without improving profit (since labor is the true bottleneck).
- Labor: fully used (1800/1800). Shadow price = 75 means each additional labor hour would increase profit by \$75, valid for up to +200 extra hours.
- Fertilizer: only 1350 units used of 2000. Slack = 650. Shadow price = 0 because it's not binding.

Economic Takeaways for Discussion

- Which resource is scarce? Labor.

- What is the operation's reservation value for an additional hour of labor capacity? \$75 per hour, before paying any cost of obtaining that capacity.
 - Why does wheat drop out of the solution? Because relative to corn it uses too much labor per profit dollar.
 - Policy thought experiment: If labor availability were increased by 200 hours, profit would rise by \$15,000 (200×75).
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3.9 Economic Interpretation

- **Shadow prices:** marginal value of resources.
 - **Allowable ranges:** robustness of those shadow prices.
 - **Managerial use:**
 - Identify which resources are most binding.
 - Assess which profit coefficients are critical.
 - Evaluate new technologies or policy changes.
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3.10 Summary

- Sensitivity analysis extends LP results beyond a single point solution.
- Provides insight into:
 - Resource valuation (RHS changes),
 - Profit robustness (objective changes),
 - Technology shifts (coefficient changes).

4 Getting Started with R & RStudio

4.1 RStudio Orientation (2–3 Minute Tour)

- **Source Pane (top-left):** where you edit scripts (`.R`) and notebooks (`.qmd`, `.Rmd`).
- **Console (bottom-left):** runs commands immediately (`>` prompt).
- **Environment/History (top-right):** objects in memory (data, vectors, functions).
- **Files/Plots/Packages/Help (bottom-right):** manage files, see plots, install packages, read docs.

Workflow tips

- Create a **Project** (File → New Project...) for the course; it pins your working directory.
 - Put scripts in a `code/` folder, data in `data/`, and outputs in `out/`.
 - Use **scripts** for anything you might need to re-run. Avoid one-off console work for assignments.
-

4.2 Preview

- Next class: **duality** — formalize the relationship between primal and dual problems.
 - Show how shadow prices emerge naturally from the dual formulation.
-

5 Duality in Linear Programming

5.1 Review

- Sensitivity analysis evaluates robustness of LP solutions to parameter changes.
- Shadow prices measure the marginal value of resources.
- Ranging analysis indicate how much parameters can change before the solution structure shifts.
- The 100% Rule helps assess simultaneous changes in parameters.
- Excel Solver provides a convenient way to solve LPs and get sensitivity reports.

5.2 Motivation for Duality

- Duality provides a **theoretical foundation** for shadow prices.
 - Every LP (the **primal**) has a corresponding **dual** LP.
 - Solutions to the dual give the **shadow prices** of the primal constraints.
 - Duality reveals deep connections between resources and values.
 - Duality also aids in sensitivity analysis and economic interpretation.
-

5.3 Canonical primal–dual pair

Primal (max form, $\leq$ constraints):

$$\begin{aligned} \max_{x \geq 0} \quad & c^\top x \\ \text{s.t.} \quad & Ax \leq b \end{aligned}$$

Dual (min form, $\geq$ constraints):

$$\begin{aligned} \min_{y \geq 0} \quad & b^\top y \\ \text{s.t.} \quad & A^\top y \geq c \end{aligned}$$

Rules of transformation (quick guide):

- One **dual variable** per **primal constraint**; one **dual constraint** per **primal variable**.
 - Primal “ $a_i^T x \leq b_i$ ” with $x \geq 0 \rightarrow$ dual variable $y_i \geq 0$. Relaxing the RHS makes the feasible set bigger, so the dual variable (shadow price) must be nonnegative.
 - Primal “ $a_i^T x \geq b_i$ ” $\rightarrow$ dual variable $y_i \leq 0$. Tightening the RHS makes the feasible set bigger (since it’s “ $\geq$ ”), so the associated price flips sign.
 - Primal “ $a_i^T x = b_i$ ” $\rightarrow$ dual variable y **free** (unrestricted in sign). An equality can cut the feasible set in either direction, so the shadow price can be positive or negative.
 - If a primal variable is **free**, the corresponding dual constraint is an **equality**; if primal variable has sign $x_j \leq 0$, flip inequality accordingly.
-

5.4 Wheat–Corn primal and its dual

Primal (wheat–corn):

$$\begin{aligned} \max \quad & 200x_W + 300x_C \\ \text{s.t.} \quad & x_W + x_C \leq 500 \quad (\text{land}) \\ & 3x_W + 4x_C \leq 1800 \quad (\text{labor}) \\ & 4x_W + 3x_C \leq 2000 \quad (\text{fertilizer}) \\ & x_W, x_C \geq 0. \end{aligned}$$

Introduce dual variables $y_1, y_2, y_3 \geq 0$ for (land, labor, fertilizer).

Dual:

$$\begin{aligned} \min \quad & 500 y_1 + 1800 y_2 + 2000 y_3 \\ \text{s.t.} \quad & y_1 + 3y_2 + 4y_3 \geq 200 \quad (\text{wheat}) \\ & y_1 + 4y_2 + 3y_3 \geq 300 \quad (\text{corn}) \\ & y_1, y_2, y_3 \geq 0 . \end{aligned}$$

Economic meanings:

y_1 = land rent (per acre), y_2 = wage (per labor hour), y_3 = fertilizer value (per unit).

5.4.1 Solve directly by intuition

A low-cost feasible choice is to try just one variable:

- Set $y_1 = 0, y_3 = 0$. The corn constraint gives $4y_2 \geq 300 \Rightarrow y_2 \geq 75$.
- Check wheat: $3y_2 \geq 200 \Rightarrow y_2 \geq 66.67$. So $y_2 = 75$ satisfies both.
- Dual objective: $500(0) + 1800(75) + 2000(0) = 135,000$.
- Could adding anything to y_1 or y_3 lower the cost?
- No, because both constraints are already satisfied and any increase would only raise the objective.
- Feasibility check:
 - Wheat: $0 + 3(75) + 4(0) = 225 \geq 200$
 - Corn: $0 + 4(75) + 3(0) = 300 \geq 300$

Let's verify in excel

5.5 Duality theorems

5.5.1 Weak duality

For any feasible solution x and y ,

$$c^\top x \leq b^\top y$$

- From the primal constraints: $Ax \leq b$
- From the dual constraints: $A^\top y \geq c$
- Premultiply the dual constraint by $x^\top \geq 0$: $x^\top A^\top y \geq x^\top c$ and note that $x^\top A^\top y = (Ax)^\top y$
- And post multiply the primal constraint by y : $(Ax)^\top y \leq b^\top y$.
- Then putting the two together gives $c^\top x \leq x^\top A^\top y = (Ax)^\top y \leq b^\top y$.

Wheat-corn example:

- Primal feasible: $x_W = 0, x_C = 450 \Rightarrow Z = 135,000$.
- Dual feasible: $y_1 = 0, y_2 = 75, y_3 = 0 \Rightarrow W = 500(0) + 1800(75) + 2000(0) = 135,000$.
- Weak duality holds: $135,000 \leq 135,000$.

5.5.2 Strong duality

If the primal and dual have optimal solutions, so does the other, and $c^\top x^* = b^\top y^*$.

- At the optimum, the primal objective (profit) equals the dual objective (resource value).
- The upper bound becomes tight: best possible activity plan = best possible valuation of resources.
- Market analogy:
 - Primal: maximize profit given resource costs.

- Dual: minimize resource costs to support given profits.
- At equilibrium, profit = cost of resources.
- Wheat-corn example:
 - Primal optimum: $Z^* = 135,000$.
 - Dual optimum: $500(0) + 1800(75) + 2000(0) = 135,000$

5.5.3 Complementary slackness conditions

The **complementary slackness conditions** link the primal and dual solutions. They provide the bridge between activity levels and resource prices:

- For each **primal constraint** $a_i^\top x \leq b_i$ with dual variable $y_i \geq 0$:

$$y_i (b_i - a_i^\top x) = 0$$

This means that either:

- The constraint is **binding** ($a_i^\top x = b_i$) and then $y_i \geq 0$,
- Or the constraint is **slack** ($a_i^\top x < b_i$) and then $y_i = 0$.
- For each **primal variable** $x_j \geq 0$ with dual inequality $a_j^\top y \geq c_j$:

$$x_j (a_j^\top y - c_j) = 0$$

This means that either:

- The activity is **produced** ($x_j > 0$) and then its dual inequality binds exactly ($a_j^\top y = c_j$),
- Or the activity is **not produced** ($x_j = 0$) and then the dual inequality can be slack ($a_j^\top y > c_j$).

5.6 Summary

- Every LP has a corresponding dual LP.
- Dual variables represent shadow prices of primal constraints.
- Weak duality: primal objective $\leq$ dual objective for any feasible solutions.
- Strong duality: at optimality, primal objective = dual objective.
- Complementary slackness links primal and dual solutions, indicating which constraints and variables are binding.
- Duality provides a theoretical foundation for sensitivity analysis and economic interpretation of LP results.

6 Extension: Input–Output Modeling

This final section is an extension of the core LP material. Input–output analysis is primarily a **linear accounting model** for tracing inter-industry linkages; its connection to LP and duality is developed at the end of the section.

6.1 Motivation

- Economies are networks of interdependent sectors.
- The output of one sector (e.g., steel) may be an input for another (e.g., car manufacturing).
- Understanding these linkages helps us analyze how a change in demand or policy ripples through the entire economy.
- Input–output (I–O) analysis was developed by **Wassily Leontief** (Nobel Prize, 1973).

6.2 Basic Input–Output Structure

- Consider an economy with n sectors.
- **Notation:**
 - x_j : total output of sector j .
 - a_{ij} : units of good i required as an intermediate input to produce one unit of good j .
 - f_j : final demand for good j (consumers, government, exports).
- **Balance equation (sector j):**

$$x_j = \sum_i a_{ij}x_j + f_j$$

- In vector notation:

$$x = Ax + f$$

where A is the matrix of technical coefficients.

Productivity Condition in Input–Output Models

For the Leontief model

$$x = Ax + f \quad \Rightarrow \quad (I - A)x = f,$$

the matrix $(I - A)$ must be invertible for a unique solution x to exist.

This requires that the **technical coefficient matrix A be productive**.

- For each sector j , the column sum

$$\sum_i a_{ij} < 1$$

means that producing one unit of output j uses **less than one unit of total inputs**.

- Leaves some output available for **final demand** ($f_j > 0$).
- If $\sum_i a_{ij} = 1$, the industry just produces to sell to other industries.
- If $\sum_i a_{ij} > 1$, the industry is not productive - uses more resources than it produces.

More formally:

The **spectral radius** (largest eigenvalue) of A satisfies $\rho(A) < 1$.

This guarantees $(I - A)^{-1}$ exists and the Leontief inverse is well-defined.

6.3 The Leontief Model

- Rearrange the balance equation:

$$x - Ax = f \Rightarrow (I - A)x = f$$

- Provided $(I - A)$ is invertible:

$$x = (I - A)^{-1}f$$

- The matrix $(I - A)^{-1}$ is called the **Leontief inverse**.
- Interpretation: it captures both the **direct** and **indirect** requirements needed to meet final demand f .
- Example: An increase in demand for cars requires more steel, which in turn requires more mining, etc.

6.4 Worked Example (3-sector economy)

Suppose three sectors: Agriculture (Ag), Manufacturing (M), and Services (S).

- **Technical coefficient matrix** the fraction of output from each sector used as input by others:
 - Rows = input sectors (what is used).
 - Columns = output sectors (what is produced).
 - Entry a_{ij} = units of input i needed per unit of output j . Ag needs 30% of its own output, 20% from M, 10% from S, etc.

$$A = \begin{bmatrix} 0.3 & 0.2 & 0.1 \\ 0.1 & 0.4 & 0.2 \\ 0.2 & 0.1 & 0.3 \end{bmatrix}$$

Input / Output	Agriculture	Manufacturing	Services
Agriculture	0.3	0.2	0.1
Manufacturing	0.1	0.4	0.2
Services	0.2	0.1	0.3

- Final demand vector:

$$f = \begin{bmatrix} 100 \\ 150 \\ 200 \end{bmatrix}$$

- Solve:

$$x = (I - A)^{-1}f$$

- Interpretation: The required outputs x will be larger than f because each sector's output must also supply intermediates to others.

```

# Technical coefficients (3x3)
A <- matrix(c(0.3,0.2,0.1,
              0.1,0.4,0.2,
              0.2,0.1,0.3), nrow = 3, byrow = TRUE)

I3 <- diag(3)

# Leontief inverse (you can also pre-specify L as below)
L <- solve(I3 - A)

# Base final demand
f <- matrix(c(100,150,200), ncol = 1)

# Compute gross outputs
x <- L %*% f
round(x, 5)

      [,1]
[1,] 336.7347
[2,] 455.1020
[3,] 446.9388

```

6.5 The Leontief Inverse and Multipliers

The matrix $(I - A)^{-1}$ is called the **Leontief inverse**.

Each element of this matrix has an important interpretation: it measures the **total requirement** of one sector's output needed to deliver one unit of final demand in another sector.

- The entry (i, j) of $(I - A)^{-1}$ tells us: "How many units of sector i are needed, directly and indirectly, to produce one additional unit of final demand for sector j ."
- **Direct requirements:** given by A .
 - Example: if $a_{12} = 0.2$, producing 1 unit of sector 2 requires 0.2 units of sector 1 directly.
- **Indirect requirements:** captured through higher powers of A .
 - A^2 shows inputs needed two steps back (e.g., steel needed to make machine tools, which then make cars).
 - A^3 shows three-step linkages, and so on.
- The series expansion makes this clear (Nuemann):

$$(I - A)^{-1} = I + A + A^2 + A^3 + \dots$$

So the Leontief inverse accumulates all **direct and indirect linkages**.

6.5.1 Multipliers

- A **multiplier** measures the total effect on the economy of a 1-unit increase in final demand for a given sector.
- In practice, the multiplier for sector j is the **column sum** of the j th column of $(I - A)^{-1}$.
 - It tells us how much total output across all sectors must rise to support 1 more unit of f_j .

- Sector-specific multipliers can also be read row by row:
 - The i, j entry shows the effect on sector i of a unit shock to demand in sector j .

6.5.2 Example (2-sector economy)

Suppose

$$A = \begin{bmatrix} 0.2 & 0.1 \\ 0.3 & 0.2 \end{bmatrix}, \quad (I - A)^{-1} = \begin{bmatrix} 1.27 & 0.16 \\ 0.48 & 1.24 \end{bmatrix}.$$

- A 1-unit increase in demand for sector 1 requires 1.27 units of sector 1 output and 0.48 units of sector 2 output in total.
- Column sums: $1.27 + 0.48 = 1.75$.
 - Interpretation: a 1-unit increase in final demand for sector 1 leads to 1.75 total units of output economy-wide (a multiplier of 1.75).
- For sector 2, the multiplier is $0.16 + 1.24 = 1.40$.

Takeaway: The Leontief inverse is not just an algebraic trick — it directly encodes the multipliers that make input–output analysis such a powerful tool for tracing the effects of shocks.

6.6 Assumptions of the Input–Output Model

- **Fixed input coefficients (Leontief technology)**
 - Each sector uses inputs in fixed proportions.
 - Example: if it takes 0.3 units of steel to make a car, every car always uses 0.3 units — no substitution is allowed.
- **Constant returns to scale**
 - Doubling output requires exactly double the inputs.
 - There are no economies or diseconomies of scale.
- **Linearity**
 - Input requirements are linear in output.
 - The balance equation $x = Ax + f$ is valid for all production levels.
- **Homogeneous outputs**
 - Each sector produces a single, uniform good.
 - All uses of that good (whether as intermediate or final demand) are treated as identical.
- **No supply constraints**
 - Any level of final demand f can, in principle, be met — resources are not limited.
 - The model solves for required gross outputs without considering feasibility of factors like labor, land, or capital.
- **Static technology and demand**
 - The input coefficients A are assumed constant over time (no innovation or efficiency change within the model run).

- The model is comparative static — it compares equilibria before and after shocks, not dynamic adjustment paths.
- **Closed vs. open model treatment**
 - In the “open” model, households are part of final demand.
 - In the “closed” model, households are included as a sector that both consumes and supplies labor.

6.7 Input–Output Analysis in Practice

The input–output (I–O) model is usually used in a **calibration + shock** framework.

Rather than solving an optimization problem, we work with observed data to trace how the economy responds to changes in demand or technology.

1. Calibration (Base Year)

- Construct the **technical coefficient matrix** A from observed data:

$$a_{ij} = \frac{\text{input of } i \text{ used by } j}{\text{total output of } j}$$

- Data typically come from national input–output tables.
- Observe base-year **final demand** f (household consumption, government, exports).
- Solve the identity

$$x = Ax + f \quad \Rightarrow \quad x = (I - A)^{-1}f$$

to check consistency with observed outputs x .

- The calibrated model reproduces the base year exactly.

2. Apply a Shock

- Change final demand f :
 - Example: +10 units of exports of manufacturing.
- Or change technical coefficients A :
 - Example: new technology reduces electricity needed in agriculture.
- Or both.

3. Recompute Outputs

- Solve again:

$$x' = (I - A)^{-1}f'$$

where f' is the new final demand (and possibly new A).

- The difference $x' - x$ shows the **direct and indirect effects** of the shock.

4. Interpret Results

- Which sectors expand the most?
- What are the **multipliers** (change in total output per unit change in final demand)?
- How do shocks propagate across the economy?

Example (illustrative):

- Suppose $f = (100, 150, 200)^\top$ and manufacturing exports rise by 20. And,

$$(I - A)^{-1} = \begin{bmatrix} 1.6 & 0.6 & 0.4 \\ 0.4 & 1.9 & 0.6 \\ 0.5 & 0.4 & 1.6 \end{bmatrix}$$

- Recalculate x' .
- Agriculture and services expand too, even though their final demand did not change — because they supply inputs to manufacturing.

```
# Shock: +20 to Manufacturing final demand
df <- matrix(c(0,20,0), ncol = 1)
f_prime <- f + df

# New outputs after shock
x_prime <- L %*% f_prime

# Change in outputs
dx <- x_prime - x

list(
  x = round(x, 5),
  x_prime = round(x_prime, 5),
  delta_x = round(dx, 5)
)
```

\$x

```
      [,1]
[1,] 336.7347
[2,] 455.1020
[3,] 446.9388
```

\$x_prime

```
      [,1]
[1,] 348.9796
[2,] 493.4694
[3,] 455.9184
```

\$delta_x

```
      [,1]
[1,] 12.24490
[2,] 38.36735
[3,]  8.97959
```

```
# Manufacturing column of L (column 2)
L_col_M <- L[,2, drop=FALSE]
check_delta <- 20 * L_col_M

cbind(
  `20 * L[,2]` = round(check_delta, 5),
  `delta x`    = round(dx, 5)
)
```

```

)

      [,1]      [,2]
[1,] 12.24490 12.24490
[2,] 38.36735 38.36735
[3,]  8.97959  8.97959

```

This “calibrate + shock” procedure is the standard way input–output analysis is used in policy and research.

6.8 Relation to LP

We can recast the input–output system as a linear program. This allows us to apply the LP tools we’ve already studied (primal, dual, shadow prices) to questions of production and pricing.

- **Primal problem (quantities).**

Decision variables are sector outputs x_j . The planner’s problem might be to:

- *Minimize cost* of meeting a fixed vector of final demands f , subject to input–output balance, or
- *Maximize welfare* given resource or technology constraints.

The balance condition $x = Ax + f$ can be rearranged as

$$Ax + f \leq x,$$

meaning that each sector’s gross output must be at least enough to cover its intermediate input requirements (Ax) plus final demand (f).

- **Objective.**

A typical choice is to minimize the total value of inputs required to satisfy demand. For example:

$$\min c^\top x$$

where c is a vector of unit costs. Alternatively, if outputs are valued in welfare terms, the objective can be to maximize $v^\top f$.

- **Constraints.**

The inequalities $Ax + f \leq x$ ensure feasibility: sectors cannot promise more intermediate and final goods than they can produce.

- **Dual problem (prices).**

The dual associates a variable p_i with each balance constraint. These p_i can be interpreted as **commodity prices** consistent with general equilibrium:

- If a commodity is scarce (constraint binding), its price is positive.
- If there is surplus, its shadow price falls to zero.

- **Strong duality.**

At the optimum,

$$\text{Value of outputs} = \text{Value of inputs}.$$

In other words, the total revenue from selling goods at equilibrium prices equals the total cost of producing them using intermediate and primary inputs. This is the familiar condition for a competitive equilibrium in input–output economics.

Draw the parallels:

- *Primal* = “quantities world”: how much each sector produces to satisfy demands.
 - *Dual* = “prices world”: what set of commodity prices makes those quantities consistent with equilibrium.
 - Duality ensures these two views are perfectly consistent.
-

6.9 Applications in Economics

- **Environmental / Resource Economics**
 - Trace carbon, water, or energy embodied in consumption.
 - Assess land-use change from shifts in demand.
 - **Policy Analysis**
 - Regional impact studies (IMPLAN, RIMS II).
 - Tariff or trade shock analysis.
 - **Research extensions**
 - Hybrid I-O and LP models for energy systems planning.
 - Coupling I-O with CGE (computable general equilibrium) for price and income effects.
-

6.10 Classroom Activity

- Provide students with a small 2×2 A matrix and demand vector f .
- Ask them to compute:
 1. $(I - A)$
 2. $(I - A)^{-1}$
 3. $x = (I - A)^{-1}f$
- Discussion prompts:
 - Which sector has the larger multiplier?
 - What happens to required outputs if one coefficient a_{ij} increases (e.g., manufacturing requires more agriculture)?